

PAYMENTGUARD PH

System Blueprint & Technical Specifications for Hybrid Anti-Fraud & Payment Verification

Document ID:	PGPH-SPEC-2026-V1	Target Market:	Philippine MSMEs & Local Retailers
Publication Date:	July 2026	Architecture:	Android Native Engine + Responsive Web Dashboard

1. EXECUTIVE SUMMARY

PaymentGuard PH is an automated, hybrid payment verification and anti-fraud ecosystem developed specifically for Micro, Small, and Medium Enterprises (MSMEs) in the Philippines. By combining an Android-native mobile application acting as an automated SMS and notification parser with a responsive cloud-backed Web Application, PaymentGuard PH provides real-time verification for digital payments made via e-wallets such as GCash, Maya, and QR Ph.

The system eliminates vulnerability to edited payment screenshots, recycled reference numbers, and communication delays between store cashiers and business owners who hold the primary SIM card off-site. Operating under a low-cost subscription model (PHP 49 to PHP 149 per month), PaymentGuard PH protects daily business revenue while streamlining store operations without requiring expensive hardware.

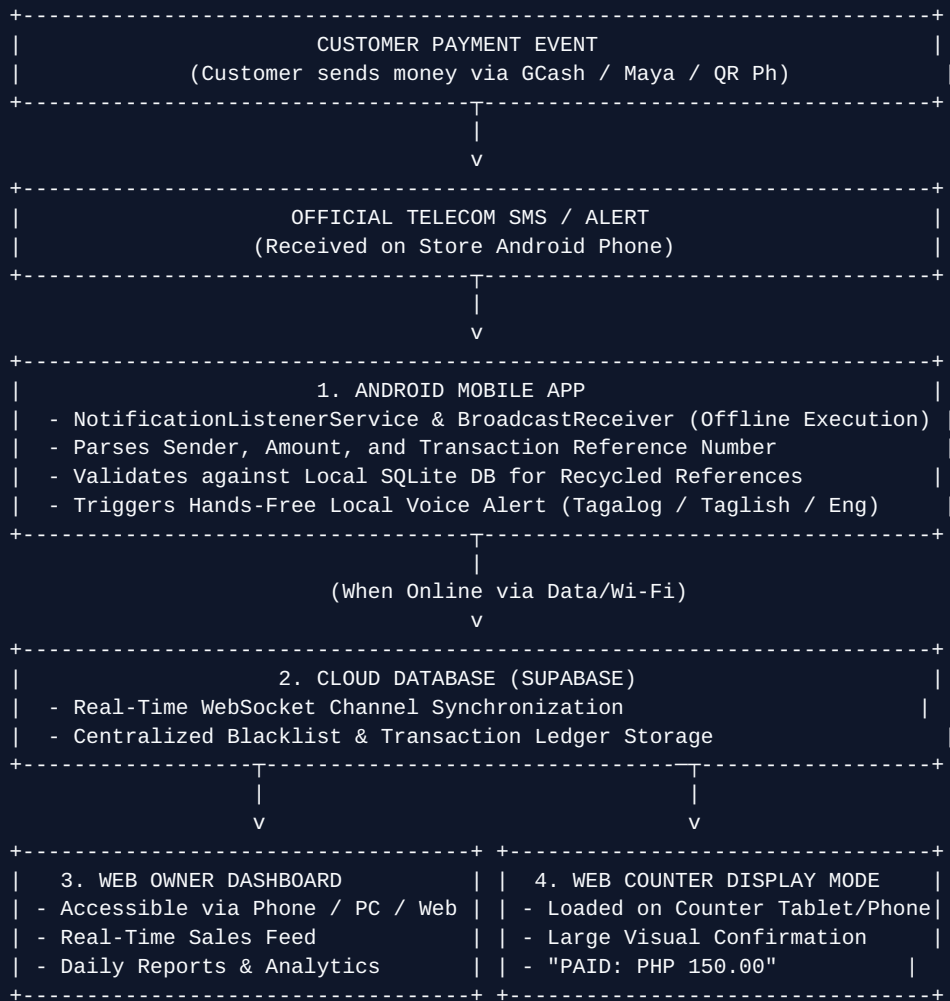
2. PROBLEM ANALYSIS & FRAUD VECTORS

Digital wallet adoption among micro-merchants in the Philippines has grown exponentially. However, existing workflows rely on cashiers manually inspecting payment confirmation screens on customer smartphones. This manual approach exposes businesses to four critical fraud vectors:

Fraud Vector / Challenge	Mechanics of Fraud	Business Impact
Fake Screen Generators	Perpetrators use mobile applications or edited image templates that render realistic confirmation screens matching GCash or Maya branding.	Direct loss of physical inventory and food goods; total revenue loss per fraudulent transaction.
Recycled Reference Numbers	Customers present authentic screenshots from legitimate transactions completed days or weeks prior at different merchant locations.	Passes superficial visual inspection by busy store staff, leading to undetected daily cash shortages.
Remote SIM Disconnect	The official registered business SIM card resides in the owner's phone at home, while hired staff manage the physical store counter.	Cashiers must call or message the owner during busy hours to confirm payment, creating bottlenecks and friction.
Intermittent Connectivity	Store locations frequently experience mobile data outages or weak cellular reception inside concrete structures.	Cloud-only verification systems freeze, forcing staff to fall back on unreliable manual screenshot inspection.

3. SYSTEM ARCHITECTURE & COMPONENT DIVISION

The PaymentGuard PH architecture employs a dual-layer strategy: a local-first Android listening service that operates independently of internet connectivity, and a cloud synchronization layer that powers a Web-based Control Dashboard and Customer Counter Display.



4. DETAILED FEATURE SPECIFICATIONS

4.1. Offline SMS & Notification Parser (Android Engine)

The core engine utilizes Android's native `NotificationListenerService` and `BroadcastReceiver` APIs to read incoming push notifications and SMS alerts emitted strictly by verified service numbers (e.g., GCash, Maya, 2882).

- **Regex Parsing Algorithm:** Extracts key data fields instantly using pattern matching: Sender ID, Transaction Amount (PHP), Timestamp, and Unique Reference Number.
- **Zero-Data Requirement:** Works entirely on local device resources. If cellular data or Wi-Fi is offline, verification and voice alerts execute without latency.

4.2. Anti-Reuse Reference Number Validator

Prevents the re-use of valid historical receipts. Every extracted Reference Number is indexed inside a local SQLite database on the Android phone.

- **Duplication Logic:** Upon reading an incoming alert, the system queries the local database for matching Reference ID strings.
- **Collision Handling:** If a matching Reference ID is detected within the database, the system immediately cancels validation and triggers a high-priority warning alert.

SYSTEM ALERT STANDARD OUTPUT — DUPLICATE DETECTED

WARNING: RECYCLED REFERENCE NUMBER DETECTED

Transaction Reference ID #4009182371 was previously processed on July 20, 2026 at 14:15:22 PST. Do not release merchandise.

4.3. Hands-Free Local Voice Alerts (Text-to-Speech)

Integrates Android's native Text-to-Speech (TTS) engine, customized for clear audio reproduction in commercial environments.

- **Language Output Options:** Tagalog, Taglish, or English.
- **Sample Speech Script:** *"Pumasok na ang isang daan at limampung piso mula kay Juan!"* (Received 150 pesos from Juan).
- **Operational Benefit:** Cashiers can remain focused on preparing food or packing items without washing hands or picking up the store mobile phone.

4.4. Web Control Dashboard for Owners

A progressive web application accessible through standard web browsers (Google Chrome, Apple Safari, Microsoft Edge) on any laptop, tablet, or smartphone.

- **Remote Monitoring:** Business owners off-site receive real-time transaction feeds as payments hit the store SIM card.
- **Sales Logging & Analytics:** Automatically aggregates daily e-wallet sales, distinguishes cash versus digital payments, and generates downloadable PDF and CSV reports.

4.5. Customer-Facing Web Counter Display Mode

A dedicated web route (`app.paymentguard.ph/counter-display`) optimized for secondary low-cost tablets or spare mobile phones positioned at the store counter facing the customer.

- **Visual Confirmation:** Receives real-time WebSocket signals from the cloud database to flash a large green status indicator: **"PAID: PHP 150.00 - THANK YOU"**.
- **Deterrence Effect:** Signals to potential bad actors that the establishment employs automated electronic verification.

4.6. Community Scammer Blacklist Network

A centralized cloud database registering reported fraudulent phone numbers and account names.

- **Incident Reporting:** Merchants can flag fraudulent attempts directly from the transaction log.
- **Network-Wide Alert:** If a blacklisted mobile number attempts a transaction at any participating PaymentGuard business, the app displays a high-risk warning immediately upon receiving the initial notification.

5. TECHNICAL STACK & IMPLEMENTATION MATRIX

Layer	Technology Selection	Implementation Rationale
Mobile App	Flutter (Dart) / Android Native Kotlin	Enables high-performance UI rendering with native Java/Kotlin method channels to hook into Android OS Notification Services.
Web Platform	React.js / Next.js or Flutter Web	Provides a lightweight, highly responsive interface for business owners across mobile and desktop browsers.
Local Storage	SQLite / Hive Database	Stores transaction reference IDs locally on the mobile device to guarantee zero-latency duplicate checks offline.
Cloud Backend	Supabase (PostgreSQL) / Firebase	Offers built-in WebSockets for instant real-time synchronization between the mobile app, cloud database, and web dashboard.

Layer	Technology Selection	Implementation Rationale
Voice Engine	Android Native TextToSpeech API	Zero operational cost; utilizes locale language packages pre-installed on Android devices.

6. BUSINESS MODEL & SUBSCRIPTION TIERS

Subscription Tier	Monthly Price	Feature Entitlements
Starter Protection	PHP 49 / month	• Android Listener Application (Single Device) • Offline Anti-Reuse Reference Number Validator • Local Tagalog/English Voice Alerts • Local Transaction History Log (7-Day Retention)
Pro Business Suite	PHP 149 / month	• All Starter Features Included • Web Owner Dashboard Access (Unlimited Remote Devices) • Customer-Facing Web Counter Display Mode • Community Scammer Blacklist Network Access • Unlimited Cloud Transaction Ledger & Exportable PDF/CSV Reports

ECONOMIC RETURN ON INVESTMENT (ROI) FOR MERCHANTS

A single prevented fraudulent transaction of PHP 150 covers the entire monthly cost of the Pro Business Suite. The product provides immediate peace of mind and operational clarity for store owners and hired staff alike.

7. IMPLEMENTATION & DEPLOYMENT ROADMAP

Phase	Timeline	Key Deliverables
Phase 1	Weeks 1 – 2	• Develop Android native service hooks for SMS and notification interception. • Build local SQLite parser and reference duplicate checking engine. • Integrate Android Text-to-Speech audio engine.
Phase 2	Weeks 3 – 4	• Establish Supabase cloud schema and real-time WebSocket channels. • Develop responsive Web Owner Dashboard and Counter Display Mode. • Implement initial Community Blacklist sync pipeline.
Phase 3	Weeks 5 – 6	• Deploy prototype across 5 pilot merchant locations (sari-sari stores, bakeries). • Conduct field testing under low-signal conditions and calibrate voice audio levels. • Refine SMS parser regex for edge cases and carrier text format variations.
Phase 4	Weeks 7 – 8	• Launch commercial onboarding via direct GCash/Maya QR subscription billing. • Deploy social proof marketing campaign showcasing live anti-fraud interception.